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Acknowledgment

You can use this chapter to give thanks to everyone who has helped bring this thesis to completion.

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Abstract

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List of Acronyms

API Application Programming Interface.

CPU Central Processing Unit.

GUI Graphical User Interface.

IP Internet Protocol.

IT Information Technology.

RAM Random Access Memory.

TCP Transmission Control Protocol.

UI User Interface.

vCPU Virtual Central Processing Unit.

VM Virtual Machine.

VPN Virtual Private Network.

List of Code Fragments

1.1	This hello-world code fragment is deceptively simple. Most rust programs are a lot more difficult to comprehend.	3
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1

Introduction

1.1 Problem Formulation

Talk about the new mindset on renewables, and that BESS is getting more and more interesting medium. <https://www.sciencedirect.com>
This paper's intro talks about renewables and its importance, and also the laws that require it. "Among all the energy storage technologies, battery technologies, especially the Li-ion battery, have experienced considerable cost reduction in the last years. Therefore, the application of Battery Energy Storage Systems (BESS) becomes a more attractive solution in electrical power systems"

Businesses often purchase their expected electricity needs on the day-ahead market. However, their actual consumption on the day of delivery can differ from what was planned. This difference, or *imbalance*, must be settled on the imbalance market. The prices in the imbalance market are significantly higher compared to the day-ahead market. Furthermore, the price is only decided at the end of a quarter, so the price for the electricity used in the past quarter will only be calculated based on the amount used during the entire quarter, multiplied by the price of the last minute of the quarter. Every minute, a forecast of the final price of the current quarter can be obtained, but this is not necessarily a good one.

To mitigate the financial risks associated with price volatility, businesses can integrate a Battery Energy Storage System (BESS) into their energy management strategy. A BESS allows a business to perform energy arbitrage by charging during periods when imbalance prices are low—or ideally negative—and discharging that energy when prices are high. This enables the business to optimize its net imbalance position for each 15-minute settlement period, substituting expensive grid intake with energy acquired at a lower cost.

The challenge is to develop an intelligent control policy for such a battery that minimizes the cost of electricity.

1 Introduction

We will discover a range of different families of techniques that can be tried for this problem.

1.2 Imbalance market

1.2.1 How it works

figure with settlement calculation maybe.

1.2.2 Why choosing that one

Potential value/loss. High risk high reward.

1.3 Related work

Acknowledgement of similarity Karimi Madahi et al but more fundamental explained, for example data split, start state configuration choices should be easier for a starter to read, and also reuse. We couldve for example chosen to use already the best one there. But because methods chosen were vague, different starting point. The ability to fill in missing information were needed gave me the decision to dive deeper in very familiar research.

1.4 This thesis

Table 1.1 shows an overview of..

Table 1.1: Overzicht resource-allocatieschema's

Naam	Jaar	Type	A F P	Invoer	Uitvoer	Getest
Alicherry et al. [1]	2012	k-sneden	A	G	par{G}	S
MCRVMP [2]	2012	ILP & GH	A	B{netwerk}	VM-plaatsing	C

Note: in most master's theses, the Introduction is a full-blow introductory chapter. That's why this chapter is numbered.

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As is visible in Figure 1.1, the cloud provider manages the IaaS and PaaS stacks.

1 Introduction

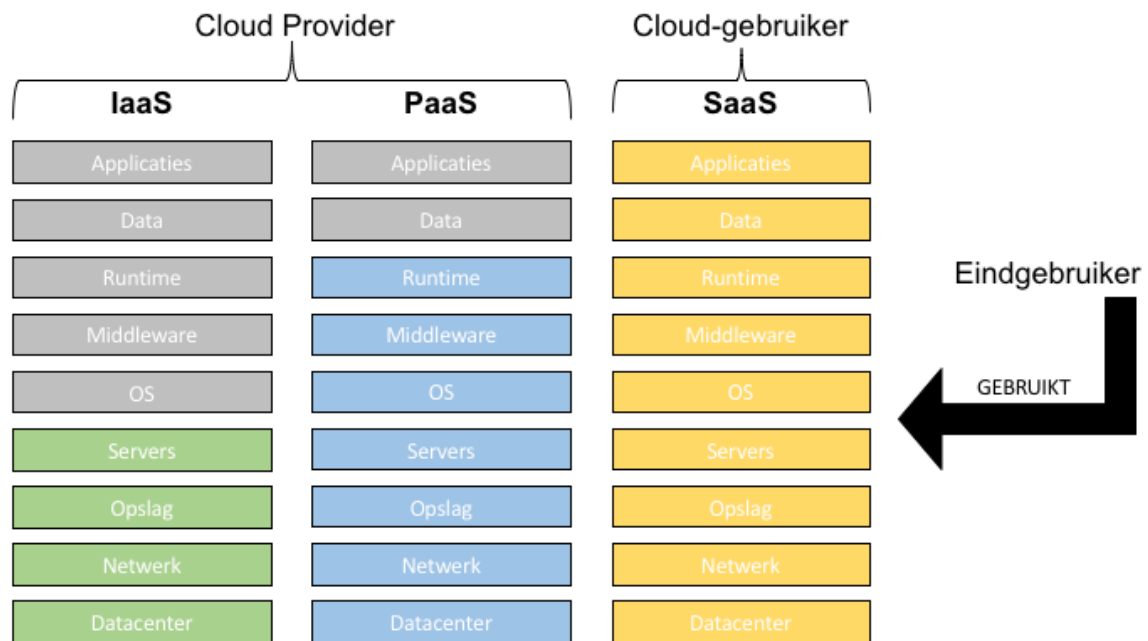


Figure 1.1: Image with caption.

Example of a code fragment. Minted supports modern programming languages such as Rust and Golang, and modern configuration formats such as YAML.

```
// This is the main function.  
fn main() {  
    // Statements here are executed when the compiled binary is called.  
  
    // Print text to the console.  
    println!("Hello World!");  
}
```

Code Fragment 1.1: This hello-world code fragment is deceptively simple. Most rust programs are a lot more difficult to comprehend.

2

Mathematical Model Formulation

This chapter serves to introduce the formal notation and state-space formulation that will be used consistently throughout this document.

2.1 Variables

In this section, the variables used in the model components are presented and clarified.

SoC_t	The amount of energy (in MWh) stored in the battery at time t .
p_t	The imbalance price (€/MWh) on the electricity grid at time t .
x_t	The amount of power (in MW) bought from the grid ($x_t > 0$) or sold to the grid ($x_t < 0$) during the interval at time t .
A_t^c, A_t^d	Accumulated energy (in MWh) traded within a 15-minute quarter. They are reset to zero at the beginning of a new quarter.
r^{chg}, r^{dis}	maximum charge/discharge rate in [MW].

2.2 Model Components

1. State Variable (S_t)

The state at time t consists of the current battery state of charge, the price of electricity at time t , which is a forecast if $(t + 1) \bmod 15 \neq 0$ given a quarter starts at time $t \bmod 15 = 0$:

$$S_t = (SoC_t, p_t, A_t^c, A_t^d)$$

2 Mathematical Model Formulation

2. Decision Variable (x_t)

The decision variable x_t represents the net energy flow to or from the battery during the time interval at time t . A positive value ($x_t > 0$) indicates charging, while a negative value ($x_t < 0$) indicates discharging. This decision is subject to the physical constraints of the battery system.

Let SoC_t be the energy stored in the battery at the beginning of the interval, SoC^{max} be the maximum battery capacity, and r^{chg} and r^{dis} be the maximum charge and discharge rates (in energy per time interval), respectively. The decision x_t must satisfy the following constraints:

$$\begin{aligned} x_t &\leq SoC^{max} - SoC_t && \text{(Cannot charge more than available capacity)} \\ x_t &\geq -SoC_t && \text{(Cannot discharge more than stored energy)} \\ x_t &\leq r^{chg} && \text{(Cannot charge faster than the maximum charge rate)} \\ x_t &\geq -r^{dis} && \text{(Cannot discharge faster than the maximum discharge rate)} \end{aligned}$$

3. Exogenous Information (W_{t+1})

This is the new, unknown information that arrives after decision x_t is made.

$$W_{t+1} = (p_{t+1})$$

4. Transition Function

$$\begin{aligned} SoC_{t+1} &= SoC_t + x_t \\ A_{t+1}^c &= \begin{cases} 0 & \text{if } (t+1) \pmod{15} = 0 \\ A_t^c + \max(0, x_t) & \text{otherwise} \end{cases} \\ A_{t+1}^d &= \begin{cases} 0 & \text{if } (t+1) \pmod{15} = 0 \\ A_t^d + \max(0, -x_t) & \text{otherwise} \end{cases} \end{aligned}$$

The price p_{t+1} is observed after the transition and is independent of the action x_t . This gives the new state vector:

$$S_{t+1} = (SoC_{t+1}, p_{t+1}, A_{t+1}^c, A_{t+1}^d)$$

5. Reward Function ($R(S_t, x_t)$)

This function defines the immediate financial outcome of an action taken at time t . Due to market rules, financial settlement occurs only at the end of each 15-minute interval. To ensure the agent learns to balance financial gains with the physical wear of the battery, the reward function calculates the *Net Profit*: the gross revenue from energy arbitrage minus the marginal degradation cost.

2 Mathematical Model Formulation

Let $E_t^{net} = A_t^c - A_t^d + x_t$ denote the net energy traded, and $E_t^{thru} = A_t^c + A_t^d + |x_t|$ denote the total energy throughput over the quarter. Let c_{wear} be the marginal degradation cost (€/MWh). The reward is defined as:

$$R(S_t, x_t) = \begin{cases} -p_t \cdot E_t^{net} - c_{wear} \cdot E_t^{thru} & \text{if } (t+1) \pmod{15} = 0 \\ 0 & \text{otherwise} \end{cases}$$

$-p_t \cdot E_t^{net}$ represents the gross financial settlement, and $c_{wear} \cdot E_t^{thru}$ represents the physical wear-and-tear cost of the energy throughput.

A positive R indicates a net profit, and a negative R indicates a net loss.

6. Objective Function

The goal of the agent is to find an optimal policy π^* that maximizes the expected cumulative discounted reward over an episode of length T . Formally, the objective function $J(\pi)$ is defined as:

$$\max_{\pi} J(\pi) = \max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t R(S_t, x_t) \right]$$

where:

- $\mathbb{E}$ denotes the expected value given the stochastic nature of the environment (prices).
- $\gamma \in [0, 1]$ is the discount factor, determining the importance of future rewards.

2.3 Battery Degradation Modeling

To accurately evaluate the profitability of an energy arbitrage agent, the physical wear of the battery system must be financially quantified. Without penalizing energy throughput, an agent would naturally execute “micro-cycles”—charging and discharging continuously for negligible price spreads—which rapidly destroys the battery’s lifespan while generating minimal revenue.

To counteract this, a constant marginal degradation cost, denoted as c_{wear} (€/MWh), is introduced into the reward function. This cost is derived using a straightforward linear depreciation model based on the total capital investment and the expected lifetime throughput of the battery:

$$c_{wear} = \frac{\text{Total Capital Cost}}{\text{Capacity} \times \text{Maximum Cycles} \times 2}$$

where the denominator calculates the total lifetime energy throughput (multiplying by two because one full cycle consists of both charging and discharging the full capacity). For example, a 2 MWh battery system with a total capital cost of €500,000 and an expected lifespan of 80,000 cycles results in a marginal throughput cost of approximately €1.56 per MWh.

2 Mathematical Model Formulation

Assumptions and Limitations

It is important to acknowledge that representing battery degradation as a constant, linear cost is a significant simplification of physical reality. In practice, lithium-ion battery degradation is a highly complex, non-linear process influenced by several dynamic operational factors:

TODO Confirm this with literature <https://www.scribd.com/document/906050666/Zhou-2011>

- **Depth of Discharge (DoD):** Deep cycles degrade the battery exponentially faster than shallow micro-cycles.
- **State of Charge (SoC):** Resting at extreme states (e.g., near 0% or 100% charge) accelerates capacity fade.
- **C-rate:** Charging or discharging at maximum power generates more heat and stress than operating at lower power levels.

Furthermore, the maximum cycle counts provided by battery manufacturers are typically determined under highly optimistic, tightly controlled laboratory conditions. These ideal benchmarks rarely reflect the harsh, variable stress induced by aggressive real-world market trading.

While incorporating an advanced, non-linear electrochemical degradation model would provide a more accurate assessment of the true trading costs, it would drastically increase the complexity of the state space and the reward landscape. Therefore, for the scope of this thesis, degradation is approximated as a linear, static cost. Integrating sophisticated battery health modeling directly into the reinforcement learning environment remains a highly relevant for future research.

2.4 State Space Decisions

In this section some extra context is given to the decisions of adding certain features.

2.4.1 A_t^c, A_t^d : Tracking Intra-Quarter Trading

The original state representation, $S_t = (SoC_t, p_t)$, was found to be insufficient for the agent to properly assign credit for its actions within a 15-minute settlement period. This created a critical ambiguity, which can be illustrated by comparing two scenarios:

- **Case 1:** The agent starts a 15-minute period with a full battery. It remains idle for 14 minutes and then discharges in the final minute. The reward for this quarter is positive, calculated as $R \approx x_{discharge} \cdot p_{15}$.
- **Case 2:** The agent starts the period with an empty battery. It charges the battery full in the first minute, holds the charge, and then discharges in the final minute. The reward for this quarter is significantly lower

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(likely negative), as it is based on the net energy traded: $R \approx (-C_{capacity} + x_{discharge}) \cdot p_{15}$.

In both cases, the state of the battery (SoC_{14}) and the action taken (x_{14}) in the final minute are identical. However, the agent receives a vastly different reward. This complicates the learning process, as the agent cannot distinguish between these two histories based solely on the current state.

To resolve this ambiguity, the state space was augmented with two new features:

- A_t^c : The total energy charged within the current quarter up to time t .
- A_t^d : The total energy discharged within the current quarter up to time t .

Justification for Two Separate Features A natural question is why two separate features are used instead of a single feature representing the net energy traded, e.g., $A_t^{net} = A_t^c - A_t^d$. While this would solve the primary ambiguity described above, it would introduce a new one. Consider two distinct scenarios:

- **Scenario A:** The agent charges for 7 minutes and then discharges for 7 minutes. The net energy traded is close to zero ($A_t^{net} \approx 0$), but the total charged and discharged volumes are significant ($A_t^c > 0$, $A_t^d > 0$).
- **Scenario B:** The agent remains idle for 14 minutes. The net energy traded is zero ($A_t^{net} = 0$), and the charged and discharged volumes are also zero ($A_t^c = 0$, $A_t^d = 0$).

With a single net feature, these two scenarios would be indistinguishable to the agent. However, Scenario A involves significant energy cycling, which is far more costly from a battery degradation perspective. By keeping A_t^c and A_t^d separate, the state representation preserves this crucial information about total energy throughput. This allows the agent to correctly correlate its actions to the final reward and provides a foundation for potentially modeling costs like battery degradation in the future.

2.5 Heuristic Policy Definition

To establish a minimum performance benchmark, a simple heuristic policy was developed. This policy relies on intuitive, transparent rules, providing a benchmark for evaluating the more complex strategies introduced later in this work.

While the general model formulation allows for any policy, the baseline heuristic employs a specific, rule-based strategy. This strategy extends the fundamental "buy-low, sell-high" principle by incorporating a trend confirmation mechanism to mitigate risk in volatile markets.

To support this trend analysis, the state observation for the heuristic is augmented with a history of the past 5

2 Mathematical Model Formulation

prices:

$$S'_t = (SoC_t, p_t, p_{t-1}, p_{t-2}, p_{t-3}, p_{t-4}, p_{t-5})$$

The policy, denoted as $X^{trend}(S'_t|\theta)$, determines the action x_t based on the current price p_t relative to tunable thresholds, and confirms the decision by checking the sign of the previous 5 price observations. The policy is defined as follows:

$$x_t = \begin{cases} r^{chg} & \text{if } p_t \leq \theta^{buy} \text{ and } p_{t-k} < 0 \text{ for all } k \in \{1, \dots, 5\} \\ r^{dis} & \text{if } p_t \geq \theta^{sell} \text{ and } p_{t-k} > 0 \text{ for all } k \in \{1, \dots, 5\} \\ 0 & \text{otherwise} \end{cases}$$

where the parameter vector to be optimized is $\theta = (\theta^{buy}, \theta^{sell})$.

This policy ensures that the agent only commits to charging when prices are consistently negative (indicating a surplus) and discharging when prices are consistently positive (indicating a shortage), thereby avoiding rapid fluctuations that could lead to unprofitable trades.

2.6 Data and Preprocessing

This section details the data acquisition, preprocessing steps, exploratory analysis, and the partitioning strategy utilized for training and evaluating the reinforcement learning agents.

2.6.1 Data Provenance and Characteristics

The primary data source for this research is Elia, the Belgian Transmission System Operator (TSO). The dataset provides electricity imbalance market data at a 1-minute resolution, accessible via Elia's OpenData platform.

A critical characteristic of this dataset is the distinction between forecasts and settlement prices. For every minute t , the published price is an indicative forecast. The definitive financial settlement price is only established at the end of each 15-minute quarter (i.e., when $t \pmod{15} = 0$). This 1-minute forecast data has only been available since the implementation of the MARI (Manually Activated Reserves Initiative) platform on May 21, 2024. Consequently, the primary dataset utilized in this research is restricted to data generated after this date.

Legacy Data Integration

While the core focus remains on post-MARI data, incorporating historical (pre-MARI) data can be theoretically valuable for expanding the training set. However, doing so requires preprocessing and harmonization. The primary

2 Mathematical Model Formulation

challenge lies in the evolution of the pricing model. The legacy format utilizes distinct columns for positive and negative imbalance prices, whereas the current format provides a single, unified imbalance price.

To integrate these datasets, legacy data must be collapsed into a single column by selecting the price corresponding to the system imbalance direction at each timestamp. Furthermore, technical naming shifts—such as the transition from "Net regulation volume" to "Area Control Error" (ACE)—require explicit feature mapping to maintain continuity in volume metrics. Finally, the feature space must be aligned by either pruning deprecated indicators (e.g., the strategic reserve price) or applying consistent padding to new analytical features (e.g., Alpha) to ensure uniform dimensionality across the entire simulation horizon.

2.6.2 Data Partitioning Principles

While analytical methods exist to solve Markov Decision Processes (MDPs) when transition probabilities are perfectly known, the complex and unpredictable nature of electricity imbalance prices renders exact mathematical solutions intractable in practice. To overcome this, this research utilizes Reinforcement Learning (RL). RL is particularly advantageous for energy arbitrage because it is a "model-free" approach; it can optimize policies in stochastic, continuous, and highly volatile environments by learning directly from interactions with historical data, entirely bypassing the need for an explicit mathematical model of the market's underlying dynamics [3].

To evaluate an RL agent's true generalization capabilities in unseen market conditions, it is fundamental to evaluate it on distinct and statistically separated datasets. In standard machine learning, the validity of such an evaluation relies heavily on the assumption that data points are Independent and Identically Distributed (i.i.d.) [4]. However, continuous time-series data from energy markets inherently violates both components of the i.i.d. assumption. Consequently, the specific methodology used to partition this data requires careful deliberation; a rigorously designed splitting strategy is essential to overcome these statistical violations and ensure that the evaluation metrics provide an as unbiased estimate of the agent's real-world performance as possible.

When partitioning time-series data, these i.i.d. violations manifest as two primary risks that must be balanced:

1. **Loss of Representativeness (Violating Identical Distribution):** A strict chronological split (e.g., reserving the final 20% of the timeline for testing) often isolates the evaluation to a specific season. Given the non-stationary nature of energy prices, a model might overfit to the test period's specific regime (e.g., winter) and fail to generalize to others. While this bias diminishes with very large, multi-year datasets where the test set encompasses several full seasonal cycles, the relatively short duration of the available post-MARI dataset (less than two years) renders chronological splitting highly unrepresentative, as the training and testing sets fail to share identical statistical distributions.
2. **Data Leakage (Violating Independence):** If data is simply shuffled randomly, the temporal sequence is broken, preventing the model from learning sequential patterns. Furthermore, because latent variables

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cause adjacent data points to be highly autocorrelated, the data points are not statistically independent. Placing day X in the training set and day $X + 1$ in the test set exploits this lack of independence, causing information leakage that results in artificially inflated evaluation scores.

To successfully navigate these risks and design an appropriate partitioning strategy, it is first necessary to conduct an exploratory data analysis. Understanding the underlying structural characteristics of the imbalance prices—such as seasonality, stationarity, and autocorrelation—dictates exactly how the data must be handled to correct these i.i.d. violations and prevent biased evaluation.

2.6.3 Exploratory Data Analysis

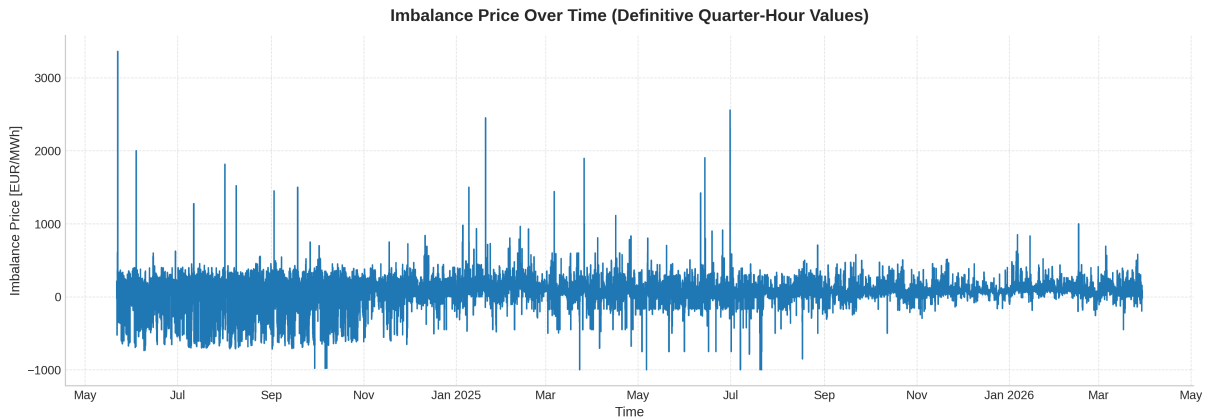


Figure 2.1: Evolution of the final quarter imbalance prices across the entire dataset.

Figure 2.1 provides a macro-level overview of the imbalance prices across the available timeline. Before proceeding to an in-depth analysis, a visual inspection of the timeline can help us identify prominent characteristics.

Most notably, the variance of the imbalance prices decreases significantly after November 2024. This drop in volatility coincides with Elia's integration into the PICASSO platform (Platform for the International Coordination of Automated Frequency Restoration and Stable System Operation). By enabling the continuous cross-border exchange of balancing energy, PICASSO increased market liquidity and significantly dampened extreme, localized price spikes within the Belgian grid.

This distinct regime shift naturally raises the question of whether the highly volatile, pre-PICASSO data remains suitable for training a model expected to operate in current, more stable market conditions. However, retaining this historical data provides valuable extra context. Exposing the model to more extreme price events expands its experiential state space, forcing it to learn risk-management and deep arbitrage (leverage) strategies.

Furthermore, the timeline shows frequent high price spikes during the summer months. In contrast, the winters of 2024-2025 and 2025-2026 remain much more stable, with almost no extreme spikes. This points to a much

2 Mathematical Model Formulation

higher variability in the summer, which is a key characteristic we will need to investigate further.

Seasonality and Stationarity

Initial analysis indicates that the imbalance market is highly non-stationary over short horizons. The statistical properties of the prices (mean, variance) change significantly depending on the season and time of day.

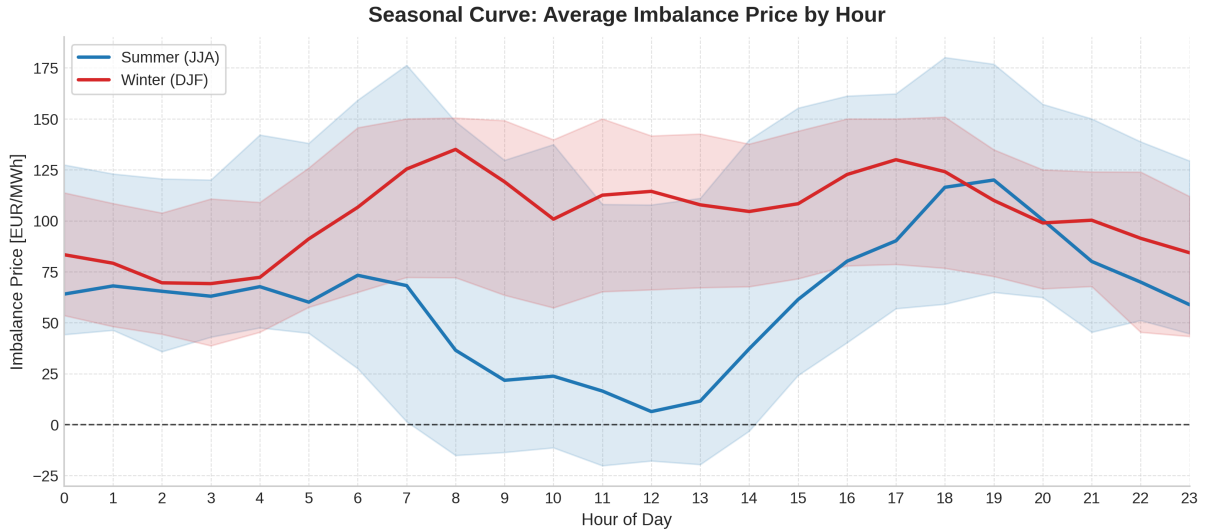


Figure 2.2: Average imbalance price profile during summer and winter months. Shaded regions represent standard deviation.

Figure 2.2 depicts the average hourly imbalance price profiles for both summer (June, July, August) and winter (December, January, February) seasons. The plot reveals two key characteristics of the electricity market:

1. **Intraday Predictability:** There exists a clear and consistent daily rhythm. Prices tend to peak in the early evening (18:00-20:00) and reach a minimum during the pre-dawn hours (03:00-05:00). This predictability enables the model to leverage anticipated intraday demand fluctuations.
2. **Inter-Seasonal Regime Shifts:** The shape of the hourly price profile is fundamentally different depending on the season. In winter, the intraday peak is more pronounced and occurs at a higher average price, whereas the summer profile is characterized by a midday dip. Furthermore, as indicated by the wider shaded regions in Figure 2.2, the standard deviation is significantly higher during the summer months compared to the winter. This increased variance reflects a higher degree of price uncertainty and volatility, likely driven by the weather-dependent nature of solar photovoltaic generation during the day.

These observations define the two fundamental requirements for the data partitioning strategy: it must preserve local temporal dependencies while simultaneously ensuring global seasonal representativeness.

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First, to preserve local temporal dependencies, the strategy must avoid the random shuffling of individual data points. Time-series data inherently exhibits *autocorrelation*—the strong statistical similarity between a given time step and its immediate predecessors. In the context of electricity markets, the price at minute t is highly dependent on the price at $t - 1$. If individual minutes are randomly assigned to different datasets, this dependency is exploited. It leads to severe data leakage, allowing the model to artificially "predict" test set values simply by recalling adjacent minutes from the training set. To prevent this, the data must be partitioned into contiguous blocks (or *episodes*). Maintaining continuous time blocks not only prevents leakage but also preserves the sequential logic required for arbitrage (e.g., the transition from a low-price charging period to a high-price discharging peak).

Second, to ensure global seasonal representativeness, these contiguous blocks cannot be drawn from just one segment of the timeline. Due to the significant regime shifts and varying volatility levels observed between summer and winter, the evaluation samples must be drawn from across the entire annual cycle. Testing the model against a diverse, year-round mixture of market conditions prevents the evaluation from being biased toward a single seasonal profile.

By combining these two principles—sampling intact, multi-day episodes from across the entire calendar year—the partitioning strategy provides a robust and mathematically sound foundation for evaluating the agent's true generalization capabilities.

An analysis of the year-long dataset, visualized in Figure 2.3, provides further insight into the nature of this time-series.

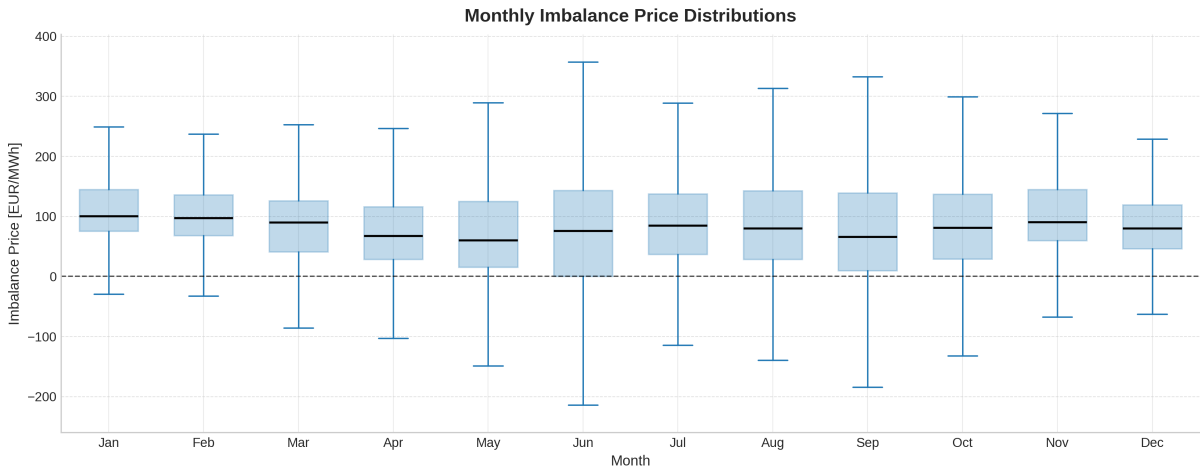


Figure 2.3: Monthly Distribution of Belgian Imbalance Prices illustrating seasonal volatility shifts.

The boxplot distributions demonstrate that while the global mean of the imbalance price remains relatively stable across the months, the variance (volatility) fluctuates significantly throughout the year. This stability in the mean suggests that the data exhibits a degree of weak-form stationarity. Because the fundamental price baseline does

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not drift into a non-stationary "random walk" (as is often the case with stock prices), partitioning the data into separate training, validation, and testing sets remains a mathematically valid approach. Provided that the sampling blocks are large enough to capture local autocorrelation, the model will encounter statistically similar price distributions in both the training and evaluation phases, allowing for a reliable assessment of its generalization capabilities.

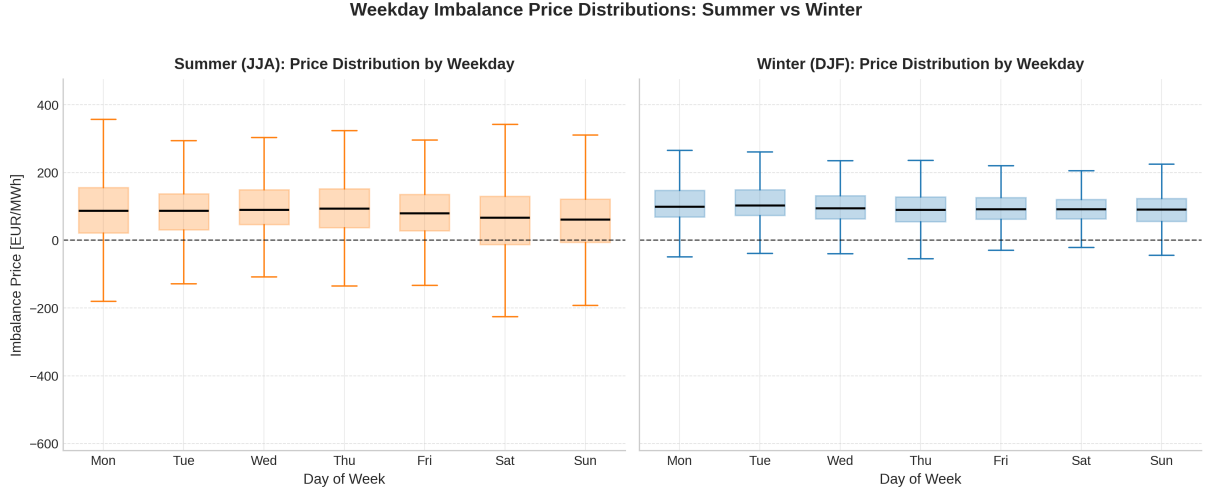


Figure 2.4: Weekly Distribution of Belgian Imbalance Prices illustrating weekly and seasonal volatility shifts.

Figure 2.4 reaffirms the contrast between winter and summer regimes, displaying a consistent mean but a distinct difference in variance. Furthermore, it demonstrates that weekly correlation is almost non-existent. Consequently, the contiguous sampling blocks do not need to span across multiple weeks to capture day-to-day dependencies, as such long-term weekly correlations barely exist in this specific market.

2.6.4 Data Partitioning: K -fold hv -Block Cross-Validation

In traditional time-series forecasting, models are typically evaluated using an out-of-sample (OOS) or "last block" validation approach, where the final chronological portion of the data is reserved for testing. However, extensive empirical studies by Bergmeir and Benítez [5] demonstrate that last-block evaluation fails to make full use of the available information and yields significantly less robust error estimates compared to cross-validation techniques. To leverage the statistical robustness of cross-validation while circumventing the theoretical violations of the independent and identically distributed (i.i.d.) assumption caused by temporal autocorrelation, they propose blocked evaluation methodologies as the superior standard procedure [6].

Building on this theoretical foundation, this research abandons the single train-test holdout approach in favor of **K -fold hv -block cross-validation**. These two concepts work in tandem: standard K -fold cross-validation provides the overarching framework for robustness by training K distinct models and comparing their performance across K independent validation and test sets [7]. However, to safely construct these K uncorrelated sets

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from continuous time-series data without causing leakage, the hv -blocking technique is utilized. This methodology systematically rotates the training and evaluation sets across the folds, maximizing data utilization [8]. The components of this approach map to established statistical theory as follows:

- **Blocked Partitioning (The v -block):** Instead of partitioning the data by individual randomized timestamps (which destroys sequence dependencies), the continuous dataset is divided into discrete, contiguous blocks of length v (e.g., multi-day episodes). As demonstrated in the exploratory data analysis 2.6.3, the electricity imbalance price time-series satisfies the condition of stationarity. This is a critical finding, as Bergmeir et al. [5] mathematically proved that blocked cross-validation yields robust, unbiased error estimates specifically under the assumption of data stationarity. With this theoretical prerequisite fulfilled, the v -blocks can be safely utilized. Furthermore, rather than keeping these blocks in strict chronological folds, they are systematically interleaved across K cross-validation folds. Distributing the v -blocks evenly across the calendar year actively ensures that every single evaluation fold contains a representative, stratified mixture of all seasonal market regimes (e.g., both high-volatility winter weeks and low-volatility summer weekends).
- **Data Purging (The h -buffer):** To explicitly prevent look-ahead bias and the leakage of autocorrelated errors, temporal gaps are introduced during every iteration of the cross-validation process. This is an application of *non-dependent* or *h -block* methodology [5]. Specifically, for any given evaluation block, a gap of size h (measured in hours) immediately preceding and following the block is “purged” (explicitly removed) from the corresponding training set. By dynamically removing this adjacent, highly correlated data at each step of the cross-validation, the evaluation sets remain rigorously isolated.

By executing the Deep Reinforcement Learning training process from scratch across all K independent folds, the resulting performance metrics can be reported with a robust variance estimate, definitively proving the stability and generalizability of the learned policies across the entire dataset.

Dataset Roles per Fold During each iteration k of the cross-validation process, the overarching dataset is dynamically partitioned into the following distinct roles:

- **Test Set:** A predefined fraction of the v -blocks assigned to fold k . This set is strictly held out and used exclusively for final performance reporting for this specific iteration.
- **Validation Set:** A second fraction of the v -blocks assigned to fold k . This set is used for hyperparameter tuning and early stopping during the training phase, ensuring configuration decisions do not bias the test results.
- **Buffer Zones:** The h -hour safety margins surrounding both the Test and Validation blocks. Data within these buffers is permanently excluded from all sets in the current fold.

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- **Training Set:** All remaining continuous chunks of data (shards) that are not claimed by the Test, Validation, or Buffer assignments.

Data Utilization and the Cost of Leakage Prevention It must be acknowledged that this approach to leakage prevention comes at a cost to data efficiency within any single training iteration. Every time a validation or test block is designated, the surrounding h -buffer hours are explicitly purged from that specific training set.

However, the overarching K -fold cross-validation framework fundamentally mitigates this limitation. In a traditional Out-Of-Sample (OOS) chronological split, the final portion of the timeline is permanently excluded from the training phase, resulting in a permanent loss of training information. In contrast, the rotational nature of K -fold cross-validation ensures that every block of data is utilized for both training and evaluation across the different iterations. Data that is purged or held out in iteration k seamlessly transitions into the training set during subsequent folds. This mechanism maximizes the extraction of information from the entire dataset over the course of the evaluation process, achieving high data utilization without compromising the strict temporal isolation required for an unbiased performance metric.

2.6.5 Fold Characteristics and Parameter Selection

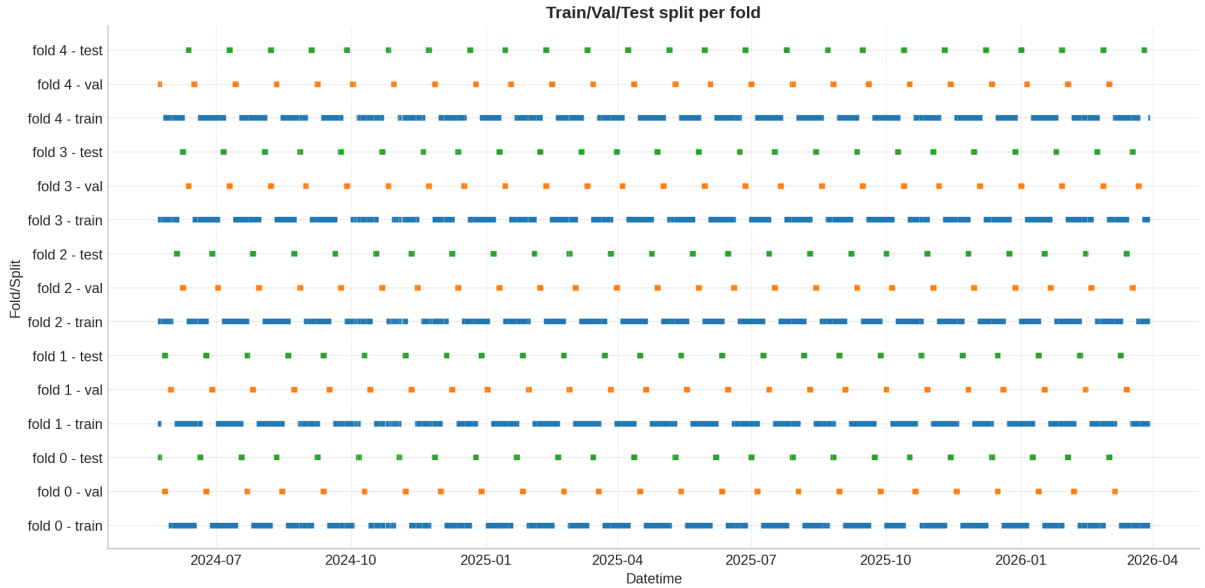


Figure 2.5: Visualization of the data partition per fold, illustrating the contiguous training blocks and the isolated, non-overlapping validation and test episodes.

This research employs a 5-fold 4 — 12-block cross-validation strategy. The choice of $K = 5$ folds aligns with standard machine learning practices, providing a sufficient number of iterations to measure the generalization capabilities of the model without having a too high computational burden.

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The block size (v) is set to 4 days, which establishes the length of a single evaluation episode. The temporal exclusion buffer (h) is set to 12 hours. This buffer size represents a necessary trade-off between maximizing data utilization and ensuring security against data leakage. Given that the total dataset spans a relatively short horizon (less than two years), the 12-hour buffer was deemed sufficient to allow temporal autocorrelations to decay while preserving as much training data as possible.

The resulting data split is visualized in Figure 2.5. The allocation algorithm strictly guarantees that there is zero overlap between the validation and test sets across the folds. Furthermore, the evaluation periods are strategically grouped to ensure that the intervening training periods remain as continuous as possible. Highly continuous training blocks are highly desirable in Reinforcement Learning, as artificial boundaries and frequent environment resets disrupt the agent’s ability to learn longer-term sequential dependencies.

To validate the efficacy of this buffered episodic splitting strategy, it is necessary to confirm that the resulting data partitions are statistically representative of one another. The sets must be independently sampled to minimize leakage, while remaining identically distributed to ensure the agent is evaluated on the same market dynamics it was trained on.

Feature Distribution Homogeneity

A core assumption of machine learning is that the training data distribution mirrors the environment the agent will encounter during deployment (the test set). If these distributions diverge significantly—a phenomenon known as covariate shift—the agent’s learned policy will fail to generalize.

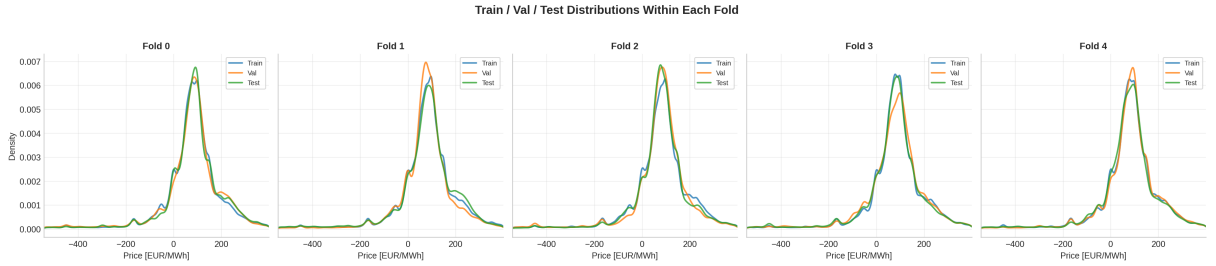


Figure 2.6: Empirical distribution of imbalance prices compared across the five distinct cross-validation folds.

As illustrated in Figure 2.6, the empirical distributions of the imbalance prices are highly consistent across all five folds. Furthermore, this consistency holds across the Training, Validation, and Testing sets within the individual folds, as seen in Figure 2.7. This homogeneity confirms that the hv -block cross-validation sampling successfully captured a representative cross-section of the market in all partitions. It provides mathematical validation that this splitting methodology successfully avoids the seasonal biases inherent in chronological splitting.

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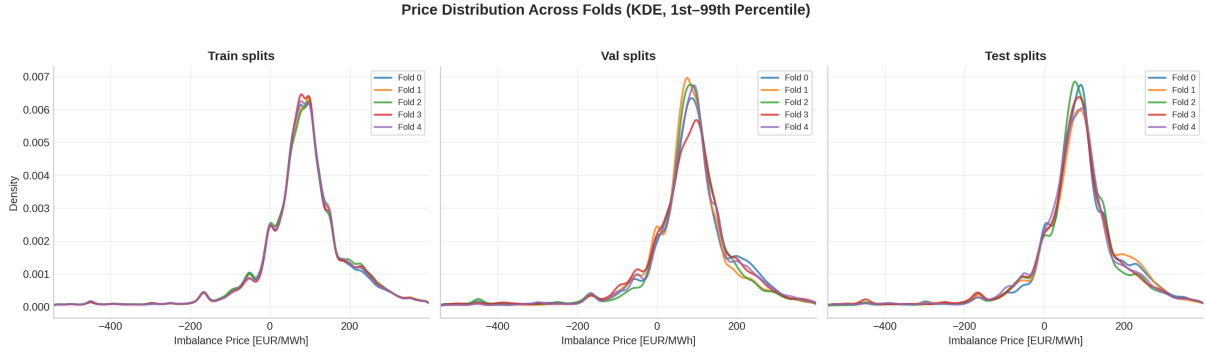


Figure 2.7: Empirical distribution of imbalance prices compared across the Training, Validation, and Testing partitions within a representative fold.

The Challenge of Latent Variables

While the visible imbalance price feature is consistently distributed, the imbalance market is heavily influenced by *latent variables*—factors that drive the price but are not explicitly provided in the agent's observation space (e.g., real-time grid frequency deviations, underlying seasonality, and intermittent renewable generation).

Because the agent cannot directly observe these underlying physical drivers, it must attempt to infer them purely from the movement of the imbalance price. This partial observability introduces an element of *irreducible error* into the learning process. The agent cannot make perfect predictions because it simply lacks the full context of the environment.

A primary focus of a subsequent chapter in this thesis will be to identify proxies for these latent variables. By engineering new features (such as time-aware encodings and historical price sequences) and systematically adding them to the state space, the objective is to reduce this irreducible error, thereby enabling the agent to learn a more optimal and profitable policy.

Data usage

This is maybe in a section after the first evaluation, so this section might need to move.

But it is an interesting question to ask if this model needs this much data, what would be too little data?

2.7 Evaluation Framework

This section establishes the general evaluation framework utilized throughout this thesis. It defines the standardized methodologies for data partitioning and performance measurement that are necessary to compare different strategies. To demonstrate the practical application of this framework, it is used here to assess the baseline

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heuristic policy. The discussion begins by detailing the evaluation criteria and concludes with an analysis of the heuristic's performance.

Evaluating the performance of an energy arbitrage agent extends beyond calculating the cumulative financial profit. A policy that yields high total returns but requires excessive battery cycling may be less desirable than a slightly less profitable but more efficient policy due to battery degradation costs. Furthermore, due to the stochastic and seasonal nature of electricity prices, a policy must be robust across different market conditions.

2.7.1 Performance Metrics

To assess the qualitative behavior and risk profile of the agent, the following metrics are computed over the test set:

1. Profitability Metrics

- **Mean Daily Reward (€/day):** The primary Key Performance Indicator (KPI). It represents the total accumulated reward divided by the total number of days in the evaluation period. This normalizes the performance regardless of the simulation length.
- **Mean Reward per Active Quarter (€/q):** The average profit generated only in quarters where the agent actively traded (non-zero reward). This serves as a proxy for efficiency. A low value indicates that the agent is "churning"—performing many trades for marginal gains—which is detrimental to battery life.

2. Reliability and Risk Metrics

- **Success Rate (Active):** The ratio of profitable quarters to total active quarters.

$$\text{Success Rate} = \frac{N_{pos}}{N_{pos} + N_{neg}}$$

where N_{pos} is the count of quarters with positive reward and N_{neg} is the count of quarters with negative reward. A high success rate implies a risk-averse, precise policy.

- **Trap Quarters (Signal Reversal Risk):** This metric quantifies how often the agent was "tricked" by intra-quarter volatility. It counts the specific instances where:

1. The agent took a position based on a favorable price at the start of the quarter ($t = 0$).
2. The price reversed direction by the settlement time ($t = 14$), resulting in a financial loss.

Formally, a "Trap Quarter" occurs if:

$$(A_{15}^c > A_{15}^d \wedge p_0 < 0 \wedge R < 0) \quad \vee \quad (A_{15}^d > A_{15}^c \wedge p_0 > 0 \wedge R < 0)$$

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A high number of trap quarters indicates that the agent fails to account for the uncertainty or volatility of the imbalance price signal.

3. Efficiency and Degradation Metrics The use of the battery is not free. In reality, Lithium-ion batteries degrade with usage; they have a finite cycle life (e.g., 5,000 cycles). A policy that generates high revenue by constantly charging and discharging for minute profit margins ("churning") may actually destroy value by wearing out the battery faster than the revenue generates a return on investment.

To evaluate this trade-off, we introduce metrics that account for the *Levelized Cost of Storage* (LCOS):

- **Daily Cycles:** This measures the physical throughput of the battery normalized by its capacity.

$$N_{cycles} = \frac{\sum_{t=0}^T |x_t|}{2 \cdot R^{max} \cdot N_{days}}$$

where $|x_t|$ is the energy throughput at time t . We divide by $2 \cdot R^{max}$ because a standard "cycle" is defined as one full charge plus one full discharge. A value significantly higher than 1.0 indicates an aggressive strategy that may accelerate degradation.

- **Profit per Cycle:** An important metric that allows us to compare different agents. It is the total profit of the of the agent, divided by the total number of cycles in a certain period.

2.8 Analysis of Results

With the mathematical formulation of the environment established, the objective shifts to identifying and evaluating optimal control policies. This analysis begins by assessing the rule-based heuristic, which serves as the foundational performance baseline.

Subsequently, we will explore deep reinforcement learning (DRL). DRL is highly suited for battery arbitrage in imbalance markets because it is a "model-free" approach. Unlike traditional stochastic optimization techniques that suffer from the curse of dimensionality, DRL algorithms can learn complex, non-linear policies directly from interaction with continuous, stochastic, and highly volatile environments, without requiring explicit prior knowledge of the market's transition probabilities [9, 10].

The initial phase of this RL exploration involves a comparative analysis of several prominent algorithms (e.g., PPO, SAC, DQN, A2C, DDPG) operating on the base environment. The objective is to identify the algorithm that achieves the highest convergence stability and cumulative financial return.

The highest-performing agent will then be selected as the exclusive algorithm for the remainder of this thesis. This is a critical methodological step, as the subsequent chapters will focus heavily on extending the state space (e.g., through advanced feature engineering) to further improve the policy's profitability.

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It is important to explicitly acknowledge a limitation of this sequential approach. The efficacy of a specific DRL algorithm is intrinsically linked to the state representation it receives. It is theoretically possible that an algorithm which underperforms on the base model might outperform the selected agent when presented with a newly engineered feature. However, due to the immense computational resources and time required to train and statistically verify multiple DRL agents from scratch, executing a full algorithmic comparison for every state-space iteration falls outside the scope of this thesis.

2.8.1 Experimental Setup

This section defines the common parameters and data used across all simulations to ensure a fair comparison between the heuristic and reinforcement learning agents.

Data and Simulation Horizon

The dataset utilized for this research spans from January 1, 2025, to December 31, 2025. All quantitative results reported hereafter are derived from the test set, which consists of 10 randomly sampled episodes, totaling 30 simulation days. To align with market settlement units, the battery system is defined with a storage capacity of 10 MWh and a power rating of 5 MW. This large-scale capacity simplifies the financial analysis, as imbalance prices are natively expressed in €/MWh.

Model Assumptions

To focus on the fundamental learning performance of the agents, several simplifying assumptions are made regarding the physical system:

- **Cycle Constraints:** No daily limit on the number of cycles is enforced. While industrial applications often require cycle limits to protect warranties [?], this study allows the agent to explore the full range of market opportunities.
- **Ideal Storage:** Round-trip efficiency is assumed to be 100% ($\eta = 1$). This means energy losses during the conversion process (heat leakage) are omitted at this stage to isolate the impact of decision logic from thermodynamic constraints.

2.8.2 Heuristic Policy Baseline

The first policy evaluated is the trend-based heuristic. The purpose of this baseline is to quantify the performance achievable through a transparent, rule-based approach before introducing neural network-based strategies.

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Parameter Selection

To establish a competitive baseline, the heuristic policy parameters were selected based on a grid search over the training dataset. The goal was to maximize the total accumulated reward while maintaining a reasonable level of risk aversion.

The selected parameters for the trend-based policy $X^{trend}(S_t|\theta)$ are:

- **Buy Threshold (θ^{buy}): 50 €/MWh.** The agent only charges when the price drops below this level, targeting clear negative-price opportunities.
- **Sell Threshold (θ^{sell}): 150 €/MWh.** The agent discharges only when prices exceed this level, ensuring a minimum spread.

Performance Evaluation

The heuristic policy was evaluated on the held-out test set using the standardized metrics defined in the evaluation framework. The quantitative results are summarized in Table 2.1. Here an active quarter is a quarter where some (dis)charging from the net is done.

Metric	Value
<i>Profitability</i>	
Mean Daily Reward	1110.23 €/day
Mean Reward per Active Quarter	62.93 €/q
Profit per Battery Cycle	1537.43 €/cycle
<i>Reliability & Activity</i>	
Profitable Quarters (N_{pos})	410
Unprofitable Quarters (N_{neg})	111
Success Rate (Active)	78.69%
Total Battery Cycles (N_{cycles})	41.12

Table 2.1: Performance Metrics for the Heuristic Baseline Policy

Discussion of Results

The heuristic achieves a commendable **Mean Daily Reward** of 1110.23 €, demonstrating that a simple rule-based approach can effectively capture the primary arbitrage opportunities in the imbalance market. The policy exhibits

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a strong **Success Rate** of 78.69%, indicating profit can quite easily be made on this market. So there is a lot of potential for more advanced heuristics/agents.

Conservatism and Absolute Thresholds As indicated by the Profit per Battery Cycle (1537.43 €), the heuristic is highly selective. While this minimizes battery wear, it also reveals a significant limitation: the policy is overly conservative. This behavior is clearly visible in Figure 2.8.

Between the 9th and 11th of January, the imbalance price remains consistently positive, fluctuating between approximately 50 €/MWh and 300 €/MWh. During this 48-hour window, the agent's State of Charge (SoC) remains at zero, and almost no trading activity occurs. Because the heuristic is bound by a fixed absolute buy threshold (θ^{buy}), it refuses to "invest" in energy at positive prices, even when clear arbitrage spreads exist.

Lack of Leverage and Arbitrage Potential This highlights the agent's inability to utilize *leverage*. In this context, leverage refers to the strategic decision to buy energy at a non-optimal (positive) price with the intent of selling it at a significantly higher price later. The heuristic lacks the foresight to recognize that a price of 80 €/MWh can be "cheap" if a spike to 400 €/MWh is imminent.

By only acting on absolute price extremes, the heuristic effectively ignores a large portion of the market's volatility. This results in the battery sitting idle during periods of high price levels where substantial revenue could be generated through relative price arbitrage. These observations justify the move toward Reinforcement Learning agents, which have the potential to learn a more dynamic policy that adapts to relative price trends rather than relying on rigid, absolute thresholds.

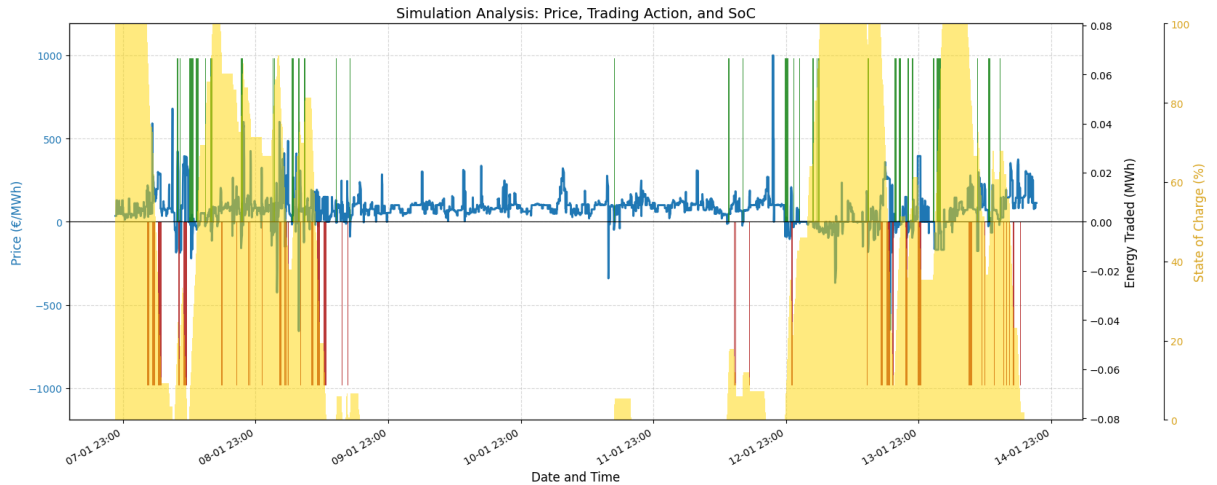


Figure 2.8: Minute-by-minute simulation trace of the heuristic policy.

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Reinforcement Learning Agents

In this section we will search for the best agent which we will use throughout the thesis. Firstly we will shortly discuss the agents and their characteristics. The test graph itself gives the mean and variance over multiple runs on the same data. !!! This should be with cross validation

2.9

3

Heuristic Policy Definition

Text of the third chapter

4

Title of the fourth chapter

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4.1 Section title 1

Text.

4.2 Section title 2

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Conclusion

Fill in..

Future Work

This chapter explains what the next steps are to continue the research and innovation of your master's thesis. Are there any additional features of research directions that are interesting? Are there ways in which your solution can be improved?

Sustainability reflection

This chapter is only for the engineering technology ("industriële ingenieur") students. It contains a sustainability reflection. You can interpret sustainability very broadly. For example, this chapter can include one or more of the following.

- Placing the work in a broader societal context. For example, how does this work impact or contribute to recent societal change such as digital transformation or the AI revolution?
- Explaining how the work contributes to the implementation of the United Nations (UN) Sustainable Development Goals (SDGs). For more information, see https://en.wikipedia.org/wiki/Sustainable_Development_Goals and <https://www.sdgs.be/nl/sdgs>.
- Reflecting on the ethical impact of this thesis or the used datasets. For example, how does this impact people's privacy? Is there bias available in the datasets? Could this be used for military or dual-use purposes?
- Investigating how the resulting work conforms to relevant laws or technical standards such as the GDPR, AI act and the CRA.

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Appendices

Attachment A

Attachment description.

Attachment B

Attachment description.